⚖️ NyayaSetu — Presentation Script (Slide-by-Slide)

Team JusticeX | 6 Members | ~8 Minutes | Live Demo Included | SIH 2026

Team Roles & Slide Assignments

Order	Speaker	Slide	Section	Time
1	**Sujal** (Team Leader)	Slide 1 — Top Half	Problem + How It Addresses the Problem	~1 min 30 sec
2	**Shraddha**	Slide 1 — Bottom Half	Detailed Solution + Innovation & Uniqueness	~1 min
3	**Sairaj**	Slide 2	Technical Approach + LIVE DEMO	~2 min
4	**Puurva**	Slide 3	Feasibility & Viability	~1 min
5	**Krishna**	Slide 4	Impact & Benefits	~1 min
6	**Nikita**	Slide 5	Research, References & SDGs	~1 min
7	**Sujal** (returns)	—	Closing Statement	~30 sec

🎬 PRE-PRESENTATION CHECKLIST

☐ Laptop charged & plugged in

☐ PPT open on Slide 1, ready to go

☐ NyayaSetu running on `localhost:3000` in a **separate browser window** (not tab — window, so you can Alt+Tab to it during Slide 2 demo)

☐ Browser has Dashboard already loaded

☐ A second tab open at `/case-status` (Public Portal)

☐ Language set to **English** (so Hindi toggle can be shown live)

☐ A test case already exists (e.g., Case #CIV/2026/0042)

☐ Screen mirrored / projector connected

☐ Clicker ready (Sujal holds it, passes to each speaker)

📊 SLIDE 1 — IDEA TITLE: न्यायसेतु - NyayaSetu

🎤 SPEAKER 1 — SUJAL (Team Leader)

Covers: "What is the Problem?" + "How It Addresses the Problem"

⏱ ~1 minute 30 seconds

> *[Sujal stands at centre. PPT shows Slide 1. He points to the pie chart on the left.]*

> **Sujal:**
>
> "Good morning, respected judges and mentors.
>
> Let me start with one number — **50 million.**
>
> That is how many court cases are pending in India right now. And as you can see on screen..."

> *[Points to the pie chart showing 55.8M, 49M, 93.143M, 29M]*

> "...this backlog is not reducing. It's growing every year. India's judicial backlog is a national crisis.
>
> But here's what most people don't realise — a huge part of this delay is not because courts lack judges. It's because of **how cases get scheduled.**
>
> Right now, court registrars prepare daily cause lists **manually** — spending 2 to 3 hours every morning assigning cases to judges and courtrooms. And the result?"

> *[Points to the bullet list under "What is the Problem?"]*

> "**Double bookings** — the same judge assigned to two hearings at once. **Courtroom clashes** — two cases in the same room at the same time. **Avoidable adjournments** — hearings cancelled because of scheduling mistakes. And when a judge falls sick? **The entire day's schedule falls apart** because there's no system to handle disruptions.

This is the problem we're solving.

Now, how does NyayaSetu address it?"

> *[Points to the right side — "How It Addresses the Problem"]*

"Our platform does five things."

> *[Reads through, pointing at each bullet]*

"**One** — It prevents scheduling conflicts by automatically detecting double-bookings, time overlaps, and unavailable resources.

Two — It reduces manual work by automating priority assessment, constraint checking, and cause-list generation.

Three — It handles disruptions — if a judge is absent or a courtroom closes, it instantly provides alternative schedules.

Four — It improves case prioritisation. Urgent cases, POCSO cases, senior citizen cases — they get heard first. Not just first-come-first-serve.

Five — And most importantly — it maintains **human control**. The staff can always Accept, Modify, or Reject every AI-assisted recommendation.

Now, let me hand over to **Shraddha** who will explain exactly how our solution works and what makes it unique."

🎤 SPEAKER 2 — SHRADDHA

Covers: "Detailed Explanation of the Proposed Solution" + "Innovation and Uniqueness"

⏱ ~1 minute

> *[Shraddha steps forward. Same Slide 1 is still on screen. She points to the bottom-left section.]*

Shraddha:

"Thank you, Sujal. Let me quickly walk you through how NyayaSetu actually works — step by step."

[Points to each bullet in "Detailed Explanation of the Proposed Solution"]

"**First — Case Priority Analysis.** When a case is registered, the system automatically scores it from 0 to 100 based on urgency, how long it's been pending, hearing duration, and special requirements like POCSO or senior citizen status.

Second — Rule-Based Scheduling. The engine assigns suitable judges, courtrooms, and available time slots — not randomly, but based on **predefined judicial rules**.

Third — Constraint Validation. Before confirming any schedule, it checks availability, double-booking, duration fit, and 6 other hard constraints. If anything fails, that option is thrown out.

Fourth — Schedule Optimisation. Among all valid options, it ranks them using priority score, judge specialisation match, workload balance, and courtroom utilisation.

Fifth — AI-Assisted Suggestions. It provides recommendations to support judges and registrars in making the final decision."

[Now points to the right side — "Innovation and Uniqueness"]

"Now — what makes us truly different from other court systems?"

[Counts on fingers]

"**Rule-Based Scheduling** — not random, not black-box. Every schedule follows predefined judicial rules and hard constraints.

Explainable Decisions — every recommendation can be verified through a Decision Receipt showing the full constraint validation.

What-If Simulation — staff can test alternative schedules safely before touching the live data.

And the most important one — **Human-in-the-Loop**. Judges and registrars always retain the final Accept, Modify, or Reject control."

[Turns toward Sairaj]

> "Now, **Sairaj** will show you all of this working live — and walk through the technical architecture."

📊 SLIDE 2 — TECHNICAL APPROACH

🎤 SPEAKER 3 — SAIRAJ

Covers: 4-Tier Architecture + Tech Stack + LIVE DEMO

⏱ ~2 minutes

> *[Sairaj is near the laptop. Slide 2 is projected.]*

> **Sairaj:**
>
> "Thank you, Shraddha. Let me first explain our architecture, and then I'll show you the actual working system."

Part A — Architecture (45 seconds)

> *[Points to the 4-Tier Architecture diagram on the left]*

> "NyayaSetu is built on a clean **4-tier architecture**.
>
> **Tier 1 — Presentation Layer.** This is what users see — separate portals for Admin, Registrar, Judge, and Public/Litigant. Built with React, TanStack Start, and Tailwind CSS.
>
> **Tier 2 — Application & Services.** This handles authentication, case management, scheduling logic, governance tracking, cause-list generation, and server functions.
>
> **Tier 3 — Intelligence & Business Logic.** This is the brain — our Smart Scheduler engine, Priority Scoring engine, What-If Digital Twin simulator, Conflict Scanner, Cause List Optimiser, and the AI Registry Copilot.
>
> **Tier 4 — Data & Persistence.** Supabase PostgreSQL with Row Level Security, storing cases, judges, courtrooms, availability, schedules, priority settings, and audit logs."

> *[Points to the Tech Stack Flow Diagram on the right]*

> "On the right, you can see our complete tech stack — React 19, TanStack Router and Query, Tailwind CSS v4, Radix UI for accessibility, Supabase for the database, Google Gemini for AI, Groq as backup, and everything deployed on Cloudflare Workers."

> *[Points to the "What-If" pipeline at the bottom]*

> "And at the bottom — our **What-If Disruption Pipeline**. When a judge goes on sick leave, the system automatically flags affected hearings, calculates the blast radius, runs a dynamic reallocation solver, shows a before-vs-after comparison, and lets the registrar approve everything with **one click**."

Part B — LIVE DEMO (1 minute 15 seconds)

> *[Alt+Tab to the browser where NyayaSetu is running]*

> "Now, let me show you this working live. Let me switch to our running application."

Step 1 — Registrar Dashboard 10 sec

> *[Dashboard is already loaded]*

> "This is the Registrar Dashboard. You can see live counters — active hearings, conflicts detected, Tier 1 cases prioritised, AI recommendations issued. All real-time from the database."

Step 2 — Smart Scheduling 25 sec

> *[Click "Smart Scheduling" in sidebar]*

> "Now let's schedule a case. I'll pick a pending case and hit Run Smart Scheduling."

> *[Click "Run Smart Scheduling" on a case]*

> "The engine just evaluated every valid Judge-Courtroom-Slot combination. Here are the top recommendations — each with a score, the assigned judge, courtroom, and time slot."

[Click on the Decision Receipt / "Why this?" button]

"And here is the Decision Receipt — you can see every hard constraint that was checked, the exact score breakdown for each soft preference, and the overall confidence. Nothing is hidden."

Step 3 — What-If Simulation 15 sec

[Navigate to "What-If Simulation"]

"What if a judge falls sick? I select a judge and a date..."

[Select a judge, click Simulate]

"Instantly — 8 hearings affected, and here are the proposed new slots for each. I review, click confirm — done. The live schedule was never at risk."

Step 4 — Hindi Toggle 10 sec

[Click the "EN | हिं" button in the top bar]

"One click..."

[Entire page switches to Hindi]

"...the entire application is now in Hindi. All navigation, all legal terms, all labels — 500+ terms translated instantly. And I can switch back just as easily."

[Click back to English]

Step 5 — Public Portal 10 sec

[Switch to the second browser tab — /case-status]

"And this is the Public Case Status portal. No login needed. A citizen types their case number..."

[Type a case number]

"...and they see their next hearing date, assigned judge, courtroom, and position in the cause list. Available in English and Hindi."

[Alt+Tab back to PPT]

"That's the system — live and working. **Puurva**, over to you for feasibility."

📊 SLIDE 3 — FEASIBILITY AND VIABILITY

🎤 SPEAKER 4 — PUURVA

Covers: Feasibility + Challenges + Strategies + Implementation Flow

⏱ ~1 minute

[Puurva steps forward. Points to the left column.]

Puurva:

"Thank you, Sairaj. Now let's talk about whether this system can actually work in real courts.

The answer is — **yes**, and here's why."

[Points to "Analysis of Feasibility"]

"**Technically feasible** — the entire system is built using deterministic, constraint-based scheduling. No complex hardware needed. Just a browser and internet.

Data-ready architecture — the system is designed around standard court entities — cases, judges, courtrooms, availability, and hearing slots. Realistic sample data can be used for initial pilot testing.

Reduces manual work — the engine automatically evaluates case priority, judge availability, courtroom availability, and time-slot constraints. What takes 2–3 hours manually, our system does in seconds."

[Points to "Potential Challenges and Risks"]

"Now, we're honest about the challenges."

"**Limited real data** — court APIs may not always be available for initial testing. **Data quality** — incomplete records can affect scheduling. **Constraint complexity** — satisfying 7 constraints simultaneously is computationally intensive. **AI reliability** — AI responses may sometimes be unsuitable."

[Points to the right — "Strategies for Overcoming"]

"But we've planned for each one."

"We use **demo data first** before connecting real court data. We run **rule checks before AI** — so AI can never recommend an invalid schedule. We keep **human control** — Accept, Modify, or Reject at every step. We show **clear reasons** for every recommendation. We maintain **audit logs** for full transparency. We support **dynamic rescheduling** when emergencies happen. And our strategy is — **start small** with a single court pilot, prove the results, then expand."

[Points to the Implementation Flow block diagram at the bottom]

"At the bottom, you can see our Implementation Flow — from data collection, to priority check, to rule-based validation, conflict checks, optimisation, recommendation, human review, final scheduling, and audit monitoring. A clean, end-to-end pipeline."

[Turns toward Krishna]

"**Krishna** will now explain the impact this creates."

📊 SLIDE 4 — IMPACT AND BENEFITS

🎤 SPEAKER 5 — KRISHNA

Covers: Impact on Target Audience + 7 Benefits + Transformation Zones

⏱ ~1 minute

[Krishna speaks with energy. Points to the three columns at the top.]

Krishna:

"Thank you, Puurva. Let's talk about who benefits from NyayaSetu and how."

[Points to the "Judges" column]

"**For Judges** — they get Explainable Recommendations with a Decision Receipt showing every constraint checked. No black box. They get a Bench View where they can directly pull urgent matters. And their final authority is always retained — they can Accept, Modify, or Reject any listing."

[Points to the "Registrars" column]

"**For Registrars** — cause-list preparation that used to take hours now takes seconds with our 3-stage batch optimiser. The system automatically scans for 8 types of conflicts before confirmation. And when disruptions happen — a judge falls sick, a courtroom closes — the What-If Digital Twin resolves it in about 60 seconds."

[Points to the "Advocates & Litigants" column]

"**For Advocates and Litigants** — statutory cases like POCSO, senior citizen matters, and long-pending cases are automatically surfaced first. Disruption risk is reduced because judge and courtroom absences are simulated before they cause impact. And the full-page Hindi translation makes the system accessible to district and taluka court staff across Hindi-speaking states."

[Points to the 7 Benefits table at the bottom]

"And here are the 7 core benefits of our solution:"

[Reads through quickly, pointing at each row]

"**Smart Case Prioritisation** — auto-ranks using 8 statutory factors.
Constraint-Aware Scheduling — guarantees zero double-booking.
Conflict Detection — flags issues before confirmation.
Workload Balancing — distributes hearings fairly across judges.
Explainable Recommendations — every choice is auditable.
What-If Simulation — tests disruptions safely.
And **Human-in-the-Loop Approval** — registrars and judges always have the final say."

[Points to the transformation zones diagram on the right]

"On the right, you can see the transformation journey — from current court challenges, through NyayaSetu's 4-step process of Prioritise, Optimise, Verify, and Adapt — to faster scheduling, fewer conflicts, better court efficiency, and complete compliance."

[Turns toward Nikita]

"**Nikita**, take us through the research."

📊 SLIDE 5 — RESEARCH AND REFERENCES

🎤 SPEAKER 6 — NIKITA

Covers: Research Conducted + References + SDG Alignment

⏱ ~1 minute

[Nikita speaks with confidence. Points to the left side.]

Nikita:

"Thank you, Krishna. Let me briefly cover the research that went into building NyayaSetu."

> *[Points to "Research Conducted" section]*

"We conducted four areas of research."

"**Judicial Backlog Analysis** — we studied India's 50+ million pending cases and the specific scheduling challenges in District and Taluka courts.

Court Scheduling Workflow — we analysed how cause lists are actually prepared manually — judge allocation, courtroom availability checks, and the adjournment patterns.

Indian Judicial Prioritisation — we researched the Government of India's mandates on POCSO, FTSC, Senior Citizen, and long-pending property case priorities. These are directly coded into our 8-factor scoring engine.

Explainable & Human-Controlled AI — we studied why black-box AI fails in judicial settings and designed our system with deterministic, rule-based scheduling where every decision is explainable and every recommendation requires human approval."

> *[Points to the "References Used" table on the right]*

"Our references include the **eCourts Mission Mode Project** by the Government of India, the **National Judicial Data Grid** for pendency and court-level statistics, the **FTSC/POCSO Scheme** by the Department of Justice, and the **POCSO Act 2012** for statutory priority considerations."

> *[Points to the three SDG cards at the bottom]*

"And our project aligns with three UN Sustainable Development Goals."

"**SDG 16 — Peace, Justice & Strong Institutions** — we strengthen judicial transparency and access to justice through smart scheduling.

SDG 9 — Industry, Innovation & Infrastructure — we introduce an innovative digital scheduling platform with explainable AI for Indian courts.

SDG 10 — Reduced Inequalities — we improve equal access to timely hearings by prioritising urgent and vulnerable cases."

> *[Turns toward Sujal]*

> "**Sujal**, take us home."

🎤 SUJAL — CLOSING (Team Leader Returns)

⏱ ~30 seconds

> *[Sujal returns to centre. Calm, direct, makes eye contact with judges.]*

> **Sujal:**
>
> "NyayaSetu — न्यायसेतु — it means **Bridge to Justice**.
>
> And that's exactly what Team JusticeX has built.
>
> A bridge from paper registers to smart scheduling.
> A bridge from black-box AI to transparent, explainable decisions.
> A bridge from 2–3 hours of manual work to cause lists generated in seconds.
> And a bridge from court delays to faster hearings for the people who need them most.
>
> 50 million cases are waiting. We've built the system to help courts start clearing them — one smart schedule at a time.
>
> Thank you."

> *[All 6 team members stand together. Ready for Q&A.]*

❓ Q&A PREPARATION — LIKELY QUESTIONS

#	Likely Question	Who	Suggested Response
1	"How is this different from eCourts or ICMS?"	**Sujal**	"eCourts digitises registers — it moves paper to screens. We go further. We add intelligent constraint-based scheduling, priority scoring, conflict detection, What-If simulation, and AI assistance — all with full transparency. eCourts doesn't schedule; we do."
2	"What if the AI gives a wrong recommendation?"	**Shraddha**	"The AI only assists — it never decides. The registrar always has the final Accept, Modify, or Reject control. And even if all AI providers fail, our scheduling engine is fully deterministic — it runs on rules, not AI."
3	"Can this handle high case volumes?"	**Sairaj**	"Yes. The engine evaluates all Judge-Courtroom-Slot combinations in memory in under 2 seconds, even with 40 judges, 20 courtrooms, and hundreds of active cases. No external compute needed."
4	"What about data privacy and security?"	**Nikita**	"Row Level Security on every database table. Role-based access control for Admin, Registrar, and Judge. AI inputs are sanitised against prompt injection. Immutable audit trails. And we support self-hosted AI models via Ollama — so court data never leaves the institution."
5	"Is the Hindi translation accurate for legal terms?"	**Krishna**	"We don't use Google Translate. We built a custom dictionary of 500+ Indian legal and judicial terms — POCSO, Section 138, bail, adjournment, cause list — all manually curated for court accuracy."
6	"How would you deploy this in a real court?"	**Puurva**	"We'd start with a single-court pilot using realistic sample data. Validate results with court staff. Then expand to more courts. The system is serverless, deployed on Cloudflare Workers, and horizontally scalable."
7	"What happens when a judge falls sick mid-day?"	**Sairaj**	"That's exactly what our What-If Simulation handles. The registrar selects the judge, clicks Simulate, and instantly sees all affected hearings with proposed new slots. She reviews, clicks

#	Likely Question	Who	Suggested Response
			Confirm, and everything is rescheduled. The live schedule was never at risk during exploration."

📋 STAGE TIPS FOR EACH SPEAKER

Speaker	Tips
Sujal	You open AND close. Speak slowly when you say "50 million" — let it sink in. Make eye contact, not the screen. During closing, slow down even more — these are your last words they'll remember.
Shraddha	You're the explainer. Use hand gestures — count the 5 solution steps on your fingers. Speak clearly, not fast. Point at the slide while talking.
Sairaj	**Practice the demo path 5 times before the presentation.** Keep clicks smooth and decisive. If something breaks or loads slowly, say: "While this loads, let me point out..." and talk about the architecture. Keep a backup screenshot tab ready as insurance.
Puurva	You handle the "can it really work?" section. Speak with confidence — especially when listing challenges. Admitting challenges shows maturity, then your strategies show preparation.
Krishna	You're the impact speaker. Use energy and enthusiasm. When listing the 7 benefits, use a rhythmic pace — like a list that builds momentum. Point at the transformation diagram with conviction.
Nikita	You're the research and credibility anchor. Speak with authority. Short sentences. Mention specific names — "Government of India", "NJDG", "POCSO Act 2012" — these show real research, not surface-level work.

⏱ TIMING SUMMARY

Section	Speaker	Duration
Slide 1 — Problem + How It Addresses	Sujal	1 min 30 sec
Slide 1 — Solution + Innovation	Shraddha	1 min
Slide 2 — Architecture + Live Demo	Sairaj	2 min
Slide 3 — Feasibility & Viability	Puurva	1 min
Slide 4 — Impact & Benefits	Krishna	1 min
Slide 5 — Research & References	Nikita	1 min
Closing	Sujal	30 sec
TOTAL		**~8 minutes**

Good luck, Team JusticeX! 🇮🇳 ⚖️